Stanley Cheung

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Education

The University of British Columbia September 2022 – May 2027
B.Sc, Major in Computer Science *Vancouver, BC*

- **Courses**: Machine Learning, Computer Networking, Relational Databases, Algorithms & Data Structures, Computer Systems & Hardware, Object-Oriented Programming
- **Awards:** 2025 Dean's List, 2024 Charles & Jane Banks Scholarship, 2023 Dean's List

Experience

Software Engineer Intern | TypeScript, Express.js, PostgreSQL, AWS, React January 2024 – July 2024
VoltSafe Inc. *Vancouver, BC*

- Designed and implemented an end-to-end RBAC system using Express.js, PostgreSQL, and AWS Cognito to secure an internal admin dashboard.
- Developed and Dockerized a server monitoring system using React, Express, and Prometheus, enabling real-time collection and analysis of CPU, RAM, and disk metrics across 7+ EC2 web and database servers.
- Built a RESTful document and image storage service for a marina management app using AWS S3, enabling users to upload, store, and retrieve files while reducing bandwidth costs.
- Conducted a performance analysis of the marina management app using Chrome Lighthouse, reducing total app bundle size by 22% by pruning unused static assets, leading to improved load times.

AP Computer Science Instructor October 2022 – February 2023
iSmart Education Canada *Vancouver, BC*

- Designed weekly algorithms, data structures, and Java lessons; supported students in achieving 5/5 on the 2023 AP CSA exam.

Projects

CreateYourStory.ai | Python, FastAPI, React, TypeScript, SQLModel | github.com/cheuyin/createyourstory.ai

- Developed and deployed a full-stack AI storytelling app using React/TypeScript and FastAPI that generates interactive branching narratives from user prompts.
- Built a resilient background-job system for asynchronous text and image generation with persisted job status, client polling, request validation, and failure handling.
- Implemented JWT/Argon2 authentication, guest sessions, ownership-based authorization, and SQLModel persistence to securely manage saved stories.

UBC Course Explorer | TypeScript, React, Express.js, Node.js | github.com/cheuyin/ubc-course-explorer

- Built a full-stack analytics app for exploring 64,000+ UBC course sections across 189 departments, featuring grade-trend charts, instructor and department profiles, and a campus room finder, using React 19, TanStack Query, and an Express + TypeScript REST API.
- Implemented a JSON query language supporting nested boolean filters (AND/OR/NOT), wildcard string matching, multi-key sorting, and GROUP-BY aggregations (MAX/MIN/AVG/SUM/COUNT), with a validation layer that returns precise field-level error messages; verified by 100+ integration tests.
- Engineered an asynchronous data-ingestion pipeline that unzips raw archives, parses course JSON, and scrapes building/room data from HTML (parse5) into a file-backed store, with job-status polling endpoints for long-running uploads and no external database.

Local AI Coding Agent | Python, Google Gemini API, Rich | github.com/cheuyin/ai-agent-python

- Engineered an autonomous agentic loop using the Google Gemini API, dispatching file system tool calls across up to 20 iterative model rounds until task completion.
- Implemented context window management that tracks prompt token usage per round and trims conversation history when approaching the 1M-token limit, enabling indefinitely long sessions.
- Blocked path traversal attacks via `os.path.commonpath` validation on all file operations; implemented 3-attempt exponential backoff retry for transient API failures.

AutoDater | TypeScript, Obsidian API | community.obsidian.md/plugins/autodater

- Launched an open-source Obsidian plugin that automates note timestamps, achieving 123 downloads and improving note traceability.

Skills

Languages: Python, TypeScript, SQL, Java, JavaScript
Frameworks: FastAPI, Express.js, React, SQLModel
Other: PostgreSQL, Docker, AWS, Linux, Git, Node.js, RESTful APIs
Certifications: AWS Certified Cloud Practitioner